Practice workbook · Compare haploid and diploid cells and explain why meiosis matters

Chromosome numbers, homologous pairs and DNA mass in body cells and gametes, karyotypes, differences from mitosis, and why halving and variation in meiosis are important.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COUNT-HAPLOID-DIPLOID-CHROMOSOMES

# 1. Question 2 · part c · 2 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c only.

**2** (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

**(1)**

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

**(1)**

(ii) State one function of mitosis in body cells.

**(1)**

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# 1. Question 2 · part c · 2 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c only.

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(c) Meiosis is another type of cell division.

Give two differences between the cells produced by meiosis and the cells produced by mitosis.

**(2)**

1 .....

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2 .....

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**(Total for Question 2 = 5 marks)**

Working and answer

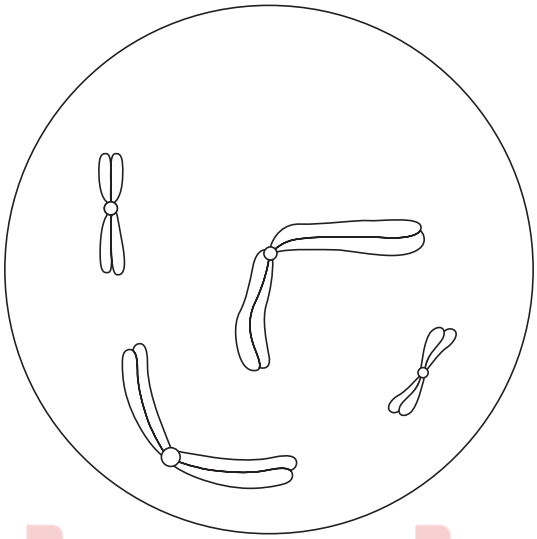
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## 2. Question 5 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

**5** The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes? **(1)**

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram. **(1)**

(iii) State how the diagram shows that this is not a human cell. **(1)**

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## 2. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

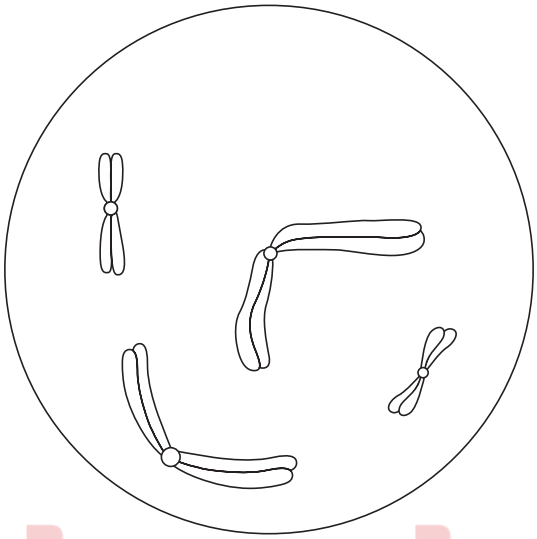
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## 3. Question 5 · part a.iii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.iii only.

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(iii) State how the diagram shows that this is not a human cell. **(1)**

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## 3. Question 5 · part a.iii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.iii only.

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(b) Describe what happens to the chromosomes during the four stages of mitosis.

**(5)**

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**(Total for Question 5 = 8 marks)**

## 3. Question 5 · part a.iii · 1 mark · Working and answer

Working and answer

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# 4. Question 5 · part c · 4 marks

January 2022 Paper 1

Further practice · Attempt part c only.

**5** (a) Describe the possible effects of a DNA mutation.

**(3)**

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

| **Age of mother in years** | 15–19 | 20–24 | 25–29 | 30–34 | 35–39 | 40–44 |
|---|---|---|---|---|---|---|
| **Probability of chromosome aberration** | 0.0010 | 0.0010 | 0.0010 | 0.0015 | 0.0045 | 0.0140 |

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P 6 7 0 5 9 A 0 1 4 2 4

# 4. Question 5 · part c · 4 marks

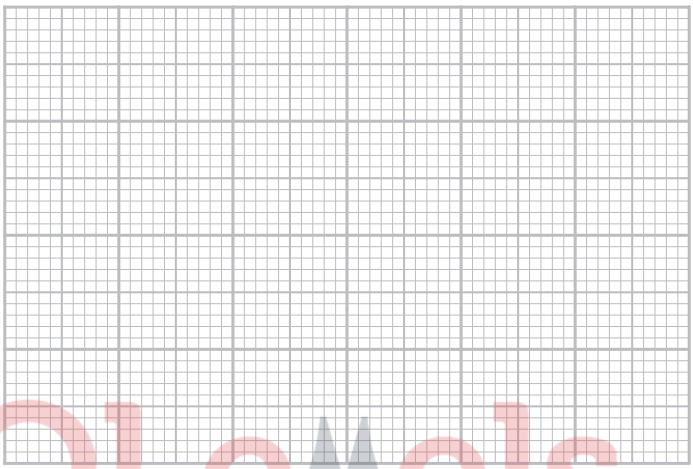
January 2022 Paper 1

Further practice · Attempt part c only.

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(i) Plot a bar chart of the results.

**(5)**



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(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

**(3)**

number of children = .....



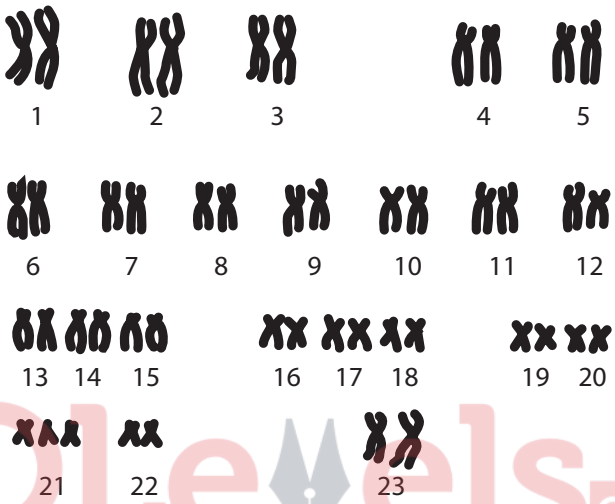
P 6 7 0 5 9 A 0 1 5 2 4

## 4. Question 5 · part c · 4 marks

January 2022 Paper 1

Further practice · Attempt part c only.

(c) The diagram shows the full diploid number of chromosomes in a person.




Justify the statement that this diagram shows the genotype of a female with a chromosome aberration.

**(4)**

**(Total for Question 5 = 15 marks)**

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## 4. Question 5 · part c · 4 marks · Working and answer

Working and answer

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# 5. Question 3 · part b.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

**3** (a) Describe what is meant by a specialised cell.

**(2)**

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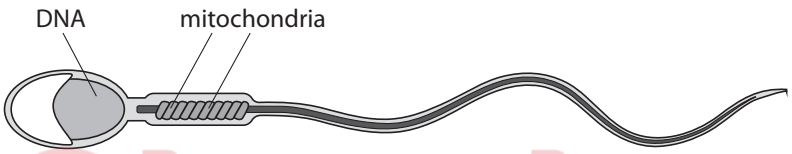
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

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(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

**(2)**

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

**(1)**

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P 7 2 5 8 8 A 0 8 2 4

## 5. Question 3 · part b.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Explain why the sperm cell has many mitochondria.

**(2)**

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

**(3)**

time taken = ..... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

**(2)**

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**(Total for Question 3 = 12 marks)**

## 5. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part b.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

**4** The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

**(1)**

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

**(1)**

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

**(3)**

| Differences | DNA | Named nucleic acid |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

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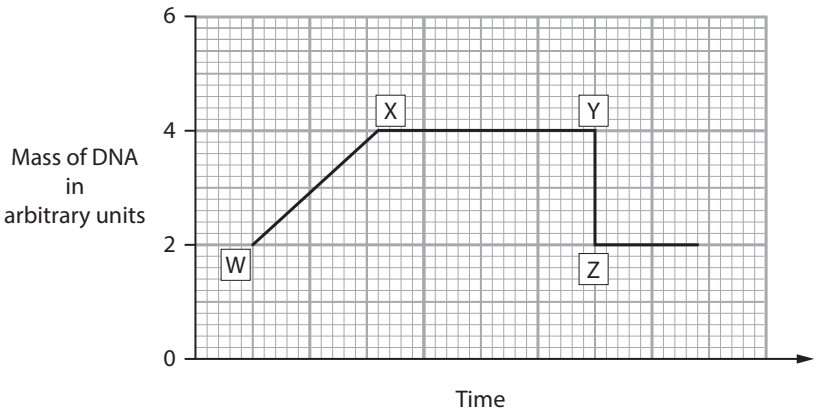
## 6. Question 4 · part b.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.




(i) Describe the process that produces the change in the mass of DNA between points W and X.

**(4)**

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(ii) Describe what is happening in the cell between points Y and Z.

**(2)**

# 6. Question 4 · part b.iii · 3 marks

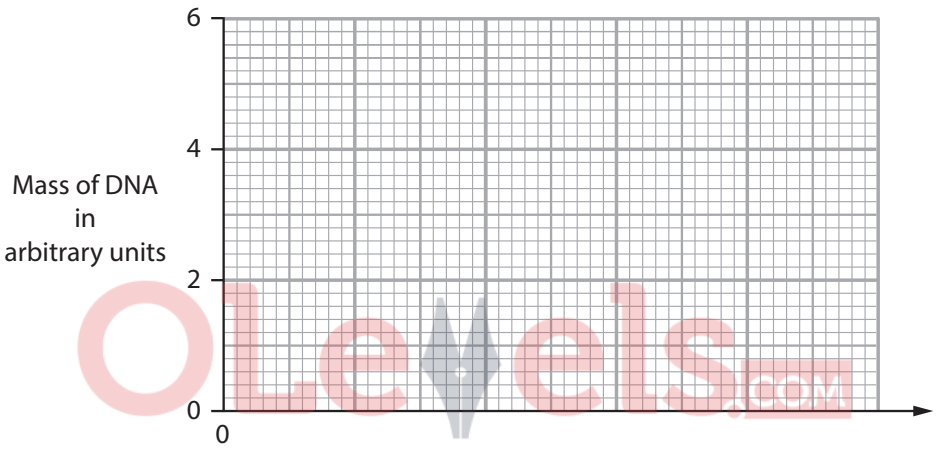
November 2023 Paper 1

Further practice · Attempt part b.iii only.

(iii) The production of gametes involves the process of meiosis.

On the grid draw a line to show the changes in the mass of DNA during meiosis.

**(3)**




**(Total for Question 4 = 14 marks)**

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Working and answer

# 7. Question 8 · part d · 3 marks

June 2019 Paper 1

Further practice · Attempt part d only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

| **Region of the world** | Q | R | S | T | U |
|---|---|---|---|---|---|
| **Total number of deaths from malaria** | 191 000 | 800 | 3800 | 14 400 | 1200 |
| **Number of deaths caused by *P. vivax*** | 1000 | 500 | 1400 | 4900 | 700 |

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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## 7. Question 8 · part d · 3 marks

June 2019 Paper 1

Further practice · Attempt part d only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

**(1)**

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

**(3)**

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

**(3)**

**(Total for Question 8 = 13 marks)**

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## 7. Question 8 · part d · 3 marks · Working and answer

Working and answer

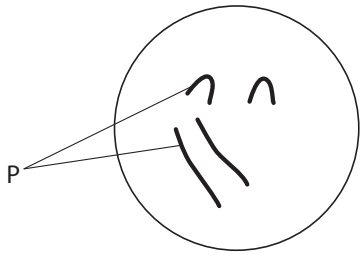
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# 8. Question 8 · part d · 4 marks

November 2020 Paper 1

Further practice · Attempt part d only.

**8** (a) The diagram shows the nucleus of a cell.



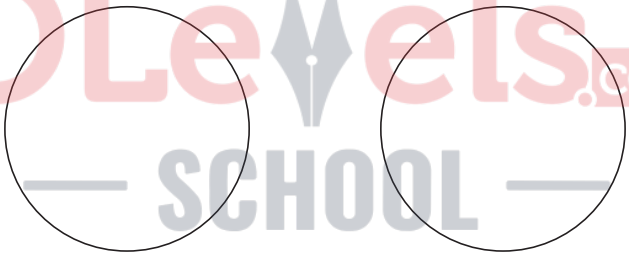

(i) Name the structures labelled P.

**(1)**

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

**(2)**



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

**(1)**

| Stage | Number |
|---|---|
| complementary bases pair up | |
| DNA double helix unwinds | |
| strands separate | |
| two DNA helices form | |

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# 8. Question 8 · part d · 4 marks

November 2020 Paper 1

Further practice · Attempt part d only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

| **Base** | X | adenine | Y | Z |
|---|---|---|---|---|
| **Pecentage base in DNA (%)** | 30 | 20 | 30 | 20 |

(i) Identify base Z.

**(1)**

(ii) Give the possible identities of base X.

**(1)**

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(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

**(4)**

**(Total for Question 8 = 10 marks)**

## 8. Question 8 · part d · 4 marks · Working and answer

Working and answer

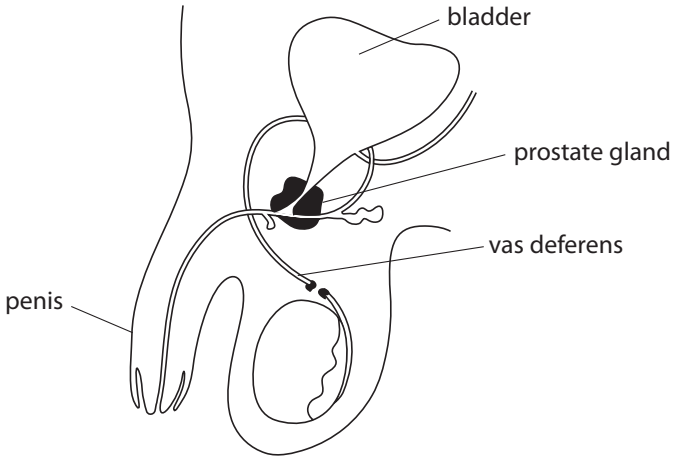
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# 9. Question 6 · part b.ii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

**6** The diagram shows the reproductive organs of a male who has had a vasectomy.




(a) (i) Explain why having a vasectomy is an effective method of contraception.

**(4)**

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## 9. Question 6 · part b.ii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(ii) Explain why this method of contraception does not prevent the spread of HIV which can cause AIDS.

**(3)**

(b) (i) Add an X to the diagram to show where haploid cells are produced.

**(1)**

(ii) Explain the importance of haploid cells in the process of fertilisation.

**(4)**

**(Total for Question 6 = 12 marks)**

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## 9. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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# 10. Question 2 · part a.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

**2** (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

**(1)**

(ii) Which of these, found in the mid-piece, release energy?

**(1)**

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

**(3)**

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## 10. Question 2 · part a.iii · 3 marks

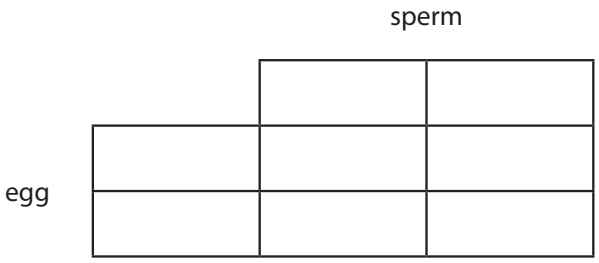
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Further practice · Attempt part a.iii only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

**(3)**




(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

**(2)**

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**(Total for Question 2 = 10 marks)**

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## 10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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# 11. Question 4 · part b.ii · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

**4** (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

| Blue | Green | Brown | Hazel | Grey |
|---|---|---|---|---|
| 𝍸 𝍸 𝍸 𝍸 𝍸 | 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 | III |

(i) Complete the table by giving the missing information.

**(1)**

| Number of people with each eye colour | | | | |
|---|---|---|---|---|
| **Blue** | **Green** | **Brown** | **Hazel** | **Grey** |
| 25 | | | | 3 |

(ii) Calculate the percentage of people with brown eyes in the sample.

**(2)**

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percentage = ..... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

**(1)**

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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## 11. Question 4 · part b.ii · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

**(4)**

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

**(2)**

(ii) Give one symptom of cystic fibrosis.

**(1)**

**(Total for Question 4 = 11 marks)**

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## 11. Question 4 · part b.ii · 4 marks · Working and answer

Working and answer

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